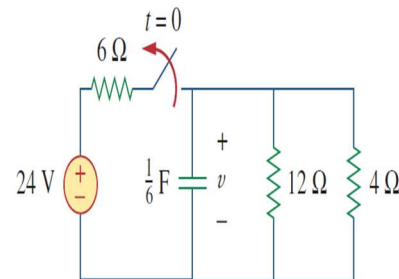


If the switch in Fig. 7.10 opens at $t = 0$, find $v(t)$ for $t \geq 0$ and $w_C(0)$.

Practice Problem 7.2

Answer: $8e^{-2t}$ V, 5.333 J.



7.3 The Source-Free RL Circuit

Figure 7.10

For Practice Prob. 7.2.

Practice Problem 7.4

For the circuit in Fig. 7.18, find $i(t)$ for $t > 0$.

Answer: $5e^{-2t}$ A, $t > 0$.

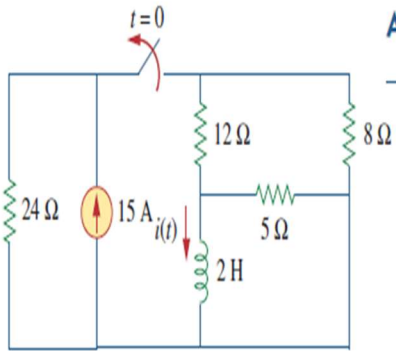


Figure 7.18

For Practice Prob. 7.4.

Determine i , i_o , and v_o for all t in the circuit shown in Fig. 7.22. Assume that the switch was closed for a long time. It should be noted that opening a switch in series with an ideal current source creates an infinite voltage at the current source terminals. Clearly this is impossible. For the purposes of problem solving, we can place a shunt resistor in parallel with the source (which now makes it a voltage source in series with a resistor). In more practical circuits, devices that act like current sources are, for the most part, electronic circuits. These circuits will allow the source to act like an ideal current source over its operating range but voltage-limit it when the load resistor becomes too large (as in an open circuit).

Answer:

$$i = \begin{cases} 16 \text{ A}, & t < 0 \\ 16e^{-2t} \text{ A}, & t \geq 0 \end{cases}, \quad i_o = \begin{cases} 8 \text{ A}, & t < 0 \\ -5.333e^{-2t} \text{ A}, & t > 0 \end{cases}$$

$$v_o = \begin{cases} 32 \text{ V}, & t < 0 \\ 10.667e^{-2t} \text{ V}, & t > 0 \end{cases}$$

Practice Problem 7.5

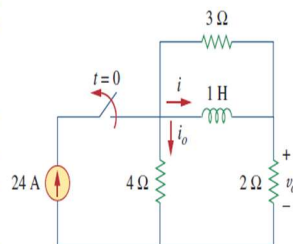


Figure 7.22

For Practice Prob. 7.5.

Example 7.10

The switch in Fig. 7.43 has been in position *A* for a long time. At $t = 0$, the switch moves to *B*. Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1$ s and 4 s.

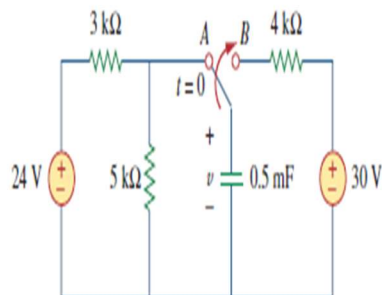


Figure 7.43

For Example 7.10.

Example 7.13

At $t = 0$, switch 1 in Fig. 7.53 is closed, and switch 2 is closed 4 s later. Find $i(t)$ for $t > 0$. Calculate i for $t = 2$ s and $t = 5$ s.

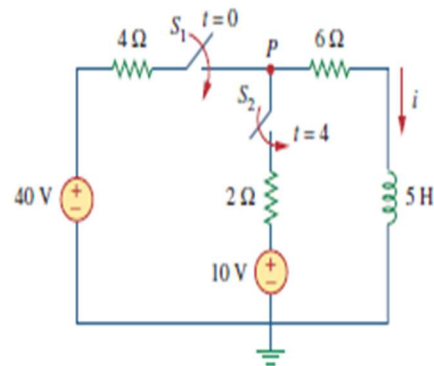


Figure 7.53

FIGURE 7.13

Practice Problem 7.17

For the circuit in Fig. 7.66, use *Pspice* to find $v(t)$ for $t > 0$.

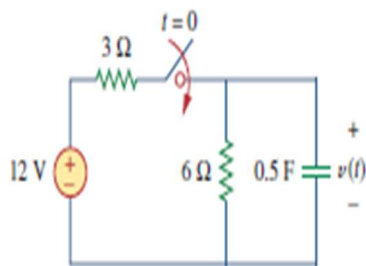


Figure 7.66

For Practice Prob. 7.17.

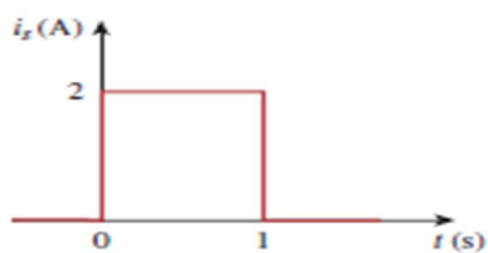
Answer: $v(t) = 8(1 - e^{-t})$ V, $t > 0$. The response is similar in shape to that in Fig. 7.65.

Practice Problem 7.21

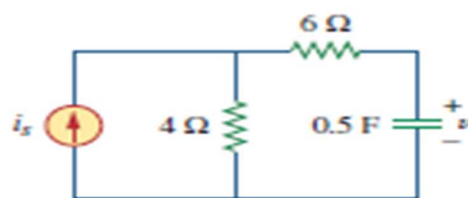
A relay has a resistance of $200\ \Omega$ and an inductance of 500 mH . The relay contacts close when the current through the coil reaches 350 mA . What time elapses between the application of 110 V to the coil and contact closure?

Answer: 2.529 ms .

7.49 If the waveform in Fig. 7.116(a) is applied to the circuit of Fig. 7.116(b), find $v(t)$. Assume $v(0) = 0$.



(a)



(b)

Figure 7.116

For Prob. 7.49 and Review Question 7.10.

7.55 Find $v(t)$ for $t < 0$ and $t > 0$ in the circuit of Fig. 7.121.

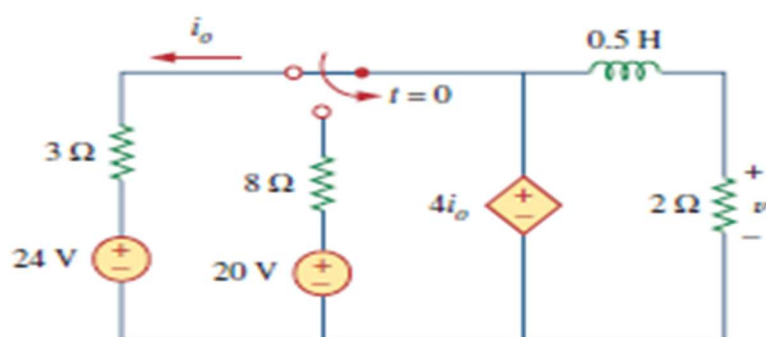


Figure 7.121
For Prob. 7.55.

the circuit.

- 7.83** An RC circuit consists of a series connection of a 120-V source, a switch, a $34\text{-M}\Omega$ resistor, and a $15\text{-}\mu\text{F}$ capacitor. The circuit is used in estimating the speed of a horse running a 4-km racetrack. The switch closes when the horse begins and opens when the horse crosses the finish line. Assuming that the capacitor charges to 85.6 V, calculate the speed of the horse.

7.87 A 120-V dc generator energizes a motor whose coil has an inductance of 50 H and a resistance of $100\ \Omega$. A field discharge resistor of $400\ \Omega$ is connected in parallel with the motor to avoid damage to the motor, as shown in Fig. 7.147. The system is at steady state. Find the current through the discharge resistor 100 ms after the breaker is tripped.

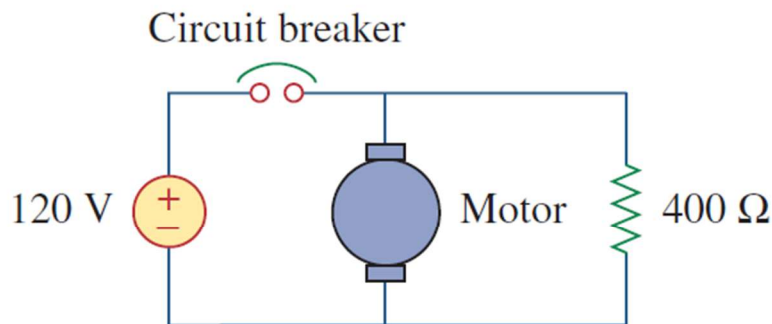


Figure 7.147
For Prob. 7.87.

7.91 The circuit in Fig. 7.150 is used by a biology student to study “frog kick.” She noticed that the frog kicked a little when the switch was closed but kicked violently for 5 s when the switch was opened. Model the frog as a resistor and calculate its resistance. Assume that it takes 10 mA for the frog to kick violently.

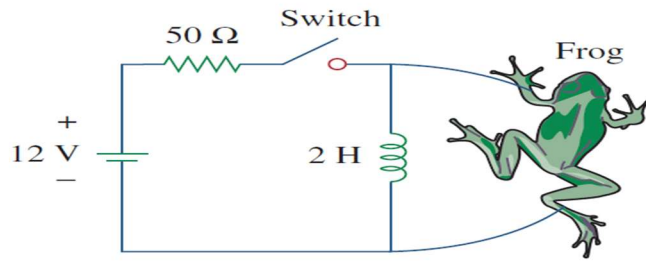
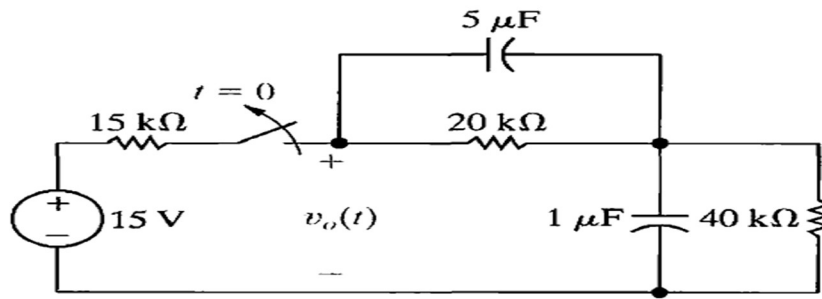


Figure 7.150
For Prob. 7.91.

7.4 The switch in the circuit shown has been closed for a long time before being opened at $t = 0$.

- Find $v_o(t)$ for $t \geq 0$.
- What percentage of the initial energy stored in the circuit has been dissipated after the switch has been open for 60 ms?



Answer: (a) $8e^{-25t} + 4e^{-10t}$ V, $t \geq 0$;

Math for Practice:

Alexander & Sadiku:

Example: 7.1, 7.2, 7.3, 7.4, 7.5, 7.10, 7.11, 7.12, 7.13, 7.17, 7.13, 7.17, 7.18, 7.21

Practice Problem: 7.1, 7.2, 7.3, 7.4, 7.5, 7.10, 7.11, 7.12, 7.13, 7.21

Exercise:

7.7, 7.8, 7.9, 7.10, 7.11, 7.13, 7.19, 7.39, 7.40, 7.42, 7.43, 7.44, 7.54, 7.55, 7.56, 7.57, 7.65, 7.83, 7.87, 7.91.

Nilsson Riedel:

Practice Problem: 7.4

Red colored: Only for final exam

Green Colored: Both for CT and Final Exam

Black Colored: Only for CT